

Spring 1994

Problem 1 (Sp86, Sp94, Sp96, Fa98). Let K be a compact subset of $\mathbb{R}^n$ and $\{B_j\}$ a sequence of open balls that covers K . Prove that there is a positive number ε such that each ε -ball centered at a point of K is contained in one of the balls B_j .

Problem 2 (Sp94). Let A be a real $n \times n$ matrix. Let M denote the maximum of the absolute values of the eigenvalues of A .

1. Prove that if A is symmetric, then $\|Ax\| \leq M\|x\|$ for all x in $\mathbb{R}^n$.
2. Prove that the preceding inequality can fail if A is not symmetric.

Problem 3 (Su77, Fa84, Sp94, Sp96, Sp09, Fa11). Evaluate the integral

$$I(a) = \int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$$

where a is real and $a > 1$. Why the formula obtained for $I(a)$ is also valid for certain complex (nonreal) values of a ?

Problem 4 (Sp94, Sp08). Let G be a group and $H \subset G$ a subgroup of finite index n . Show that G contains a normal subgroup N such that $N \subset H$ and the index of N is at most $n!$.

Problem 5 (Sp94). 1. Suppose the functions $\sin t$ and $\sin 2t$ are both solutions of the differential equation

$$\sum_{k=0}^n c_k \frac{d^k x}{dt^k} = 0$$

where $c_0, \dots, c_n$ are real constants. What is the smallest possible order of the equation? Write down an equation of minimum order having the given functions as solutions.

2. Will the answers to Part 1 be different if the constants $c_0, \dots, c_n$ are allowed to be complex?

Problem 6 (Sp94). Prove or disprove: A square complex matrix, A , is similar to its transpose, A^T .

Problem 7 (Sp94). Let $a_1, a_2, \dots, a_n$ be complex numbers. Prove that there is a point x in $[0, 1]$ such that

$$\left| 1 - \sum_{k=1}^n a_k e^{2\pi i k x} \right| \geq 1$$

Problem 8 (Sp94). Find all automorphisms of $\mathbb{Z}[x]$, the ring of polynomials over $\mathbb{Z}$.

Problem 9 (Sp94). 1. Let U and V be open connected subsets of the complex plane, and let f be an analytic function in U such that $f(U) \subset V$. Assume $f^{-1}(K)$ is compact whenever K is a compact subset of V . Prove that $f(U) = V$.

2. Prove that the last equality can fail if *analytic* is replaced by *continuous* in the preceding statement.

Problem 10 (Sp94). Let f be a continuous real valued function on $\mathbb{R}$ such that the improper Riemann integral $\int_{-\infty}^{\infty} |f(x)| dx$ converges. Define the function g on $\mathbb{R}$ by

$$g(y) = \int_{-\infty}^{\infty} f(x) \cos xy \, dx$$

Prove that g is continuous.

Problem 11 (Sp94). Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a diagonalizable linear transformation. Prove that there is an orthonormal basis for $\mathbb{R}^n$ with respect to which T has an upper triangular matrix.

Problem 12 (Sp94). Let $f = u + iv$ and $g = p + iq$ be analytic functions defined in a neighborhood of the origin in the complex plane. Assume $|g'(0)| < |f'(0)|$. Prove that there is a neighborhood of the origin in which the function $h = f + \bar{g}$ is one-to-one.

Problem 13 (Sp94). Let $\mathbb{F}_q$ be a finite field with q elements. Say that a function $f : \mathbb{F}_q \rightarrow \mathbb{F}_q$ is a *polynomial function* if there are elements $a_0, a_1, \dots, a_n$ of $\mathbb{F}_q$ such that $f(x) = a_0 + a_1x + \dots + a_nx^n$ for all $x \in \mathbb{F}_q$. How many polynomial functions are there?

Problem 14 (Sp94). Let W be a real 3×3 antisymmetric matrix, i.e., $W^T = -W$. Let the function

$$X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix}$$

be a real solution of the vector differential equation $\frac{dX}{dt} = WX$.

1. Prove that $\|X(t)\|$, the Euclidean norm of $X(t)$, is independent of t .
2. Prove that if v is a vector in the null space of W , then $X(t) \cdot v$ is independent of t .
3. Prove that the values $X(t)$ all lie on a fixed circle in $\mathbb{R}^3$.

Problem 15 (Fa89, Sp94). Let α be a number in $(0, 1)$. Prove that any sequence (x_n) of real numbers satisfying the recurrence relation

$$x_{n+1} = \alpha x_n + (1 - \alpha)x_{n-1}$$

has a limit, and find an expression for the limit in terms of α , x_0 and x_1 .

Problem 16 (Sp94). For which numbers a in $(1, \infty)$ is it true that $x^a \leq a^x$ for all x in $(1, \infty)$?

Problem 17 (Sp90, Fa93, Sp94). Show that there are at least two nonisomorphic non-abelian groups of order 24, of order 30 and order 40.

Problem 18 (Sp94). Let u be a real valued harmonic function in the complex plane such that

$$u(z) \leq a |\log |z|| + b$$

for all z , where a and b are positive constants. Prove that u is constant.